

# ECO101 - Introduction to Microeconomics

Department of Economics and Social Sciences (ESS), Brac University

# Short run vs. long run total, marginal, and average costs

## Lecture: 6

# Reading

- Economics by Michael Parkin, 10th edition
  - Chapter: 11
- Principles of Economics by Mankiw, 6th Edition
  - Chapter: 13

# Outline

- Explain the relationship between a firm's output and costs in the short run
- Derive a firm's short-run cost curves
- Explain the relationship between a firm's output and costs in the long run
- Derive a firm's long-run average cost curve

# Introduction

- Every firm must decide how much to produce, how many people to employ, and how much and what type of capital equipment to use
- How firms cope with its economic problems and makes decisions related to the production process?

This lecture will give us a clear view of the problems faced by all firms in the real world

# Short-Run Cost

To produce more output in the short run, a firm must employ more labour, which means that it must increase its costs

The relationship between output and cost is described by using three cost concepts:

- Total cost
- Marginal cost
- Average cost

# Total Cost

- A firm's **total cost (TC)** is the cost of all the factors of production it uses



- **Total fixed cost (TFC)** is the cost of the firm's fixed factors. The quantities of fixed factors do not change as output changes, so total fixed cost is the same at all outputs. For a knitting firm, total fixed cost includes the cost of renting knitting machines
- **Total variable cost (TVC)** is the cost of the firm's variable factors. Total variable cost changes as output changes. For the knitting firm, labour is the variable factor, so this component of cost is its wage bill
- Total cost is the sum of total fixed cost and total variable cost

$$TC = TFC + TVC.$$

# Total Cost

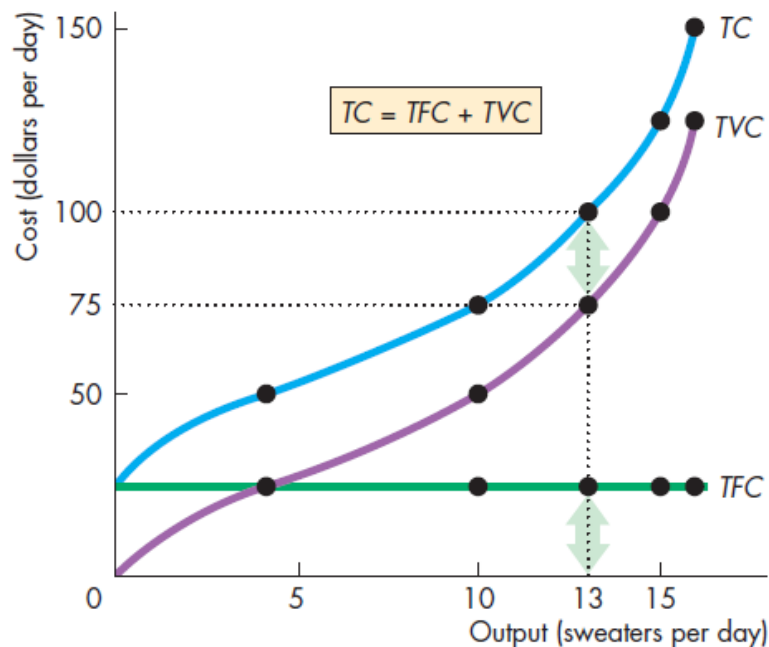
|          |                               |                                 | Total<br>fixed<br>cost<br>(TFC) | Total<br>variable<br>cost<br>(TVC) | Total<br>cost<br>(TC) |
|----------|-------------------------------|---------------------------------|---------------------------------|------------------------------------|-----------------------|
|          | Labor<br>(workers<br>per day) | Output<br>(sweaters<br>per day) | (dollars per day)               |                                    |                       |
| A        | 0                             | 0                               | 25                              | 0                                  | 25                    |
| B        | 1                             | 4                               | 25                              | 25                                 | 50                    |
| C        | 2                             | 10                              | 25                              | 50                                 | 75                    |
| <b>D</b> | <b>3</b>                      | <b>13</b>                       | <b>25</b>                       | <b>75</b>                          | <b>100</b>            |
| E        | 4                             | 15                              | 25                              | 100                                | 125                   |
| F        | 5                             | 16                              | 25                              | 125                                | 150                   |

Firm rents knitting machine for \$25 a day and hires workers at a wage rate of \$25 a day

In row D,  
 $TVC = 3 \times \$25 = \$75$   
 $TC = TFC + TVC$   
 $= \$25 + \$75 = \$100$

# Total Cost Curves

**FIGURE 11.4** Total Cost Curves



- Total fixed cost is constant
- Total variable cost increases as output increases, so the TVC curve and the TC curve increase as output increases
- The vertical distance between the TC curve and the TVC curve equals TFC (total fixed cost)

# Marginal Cost

- A firm's **marginal cost** is the increase in total cost that results from a one-unit increase in output
- Marginal cost is calculated as the increase in total cost divided by the increase in output

$$MC = \Delta TC \div \Delta Q$$

- For example, if output increases from 10 sweaters to 13 sweaters and total cost increases from \$75 to \$100, then the change in output is 3 sweaters, and the change in total cost is \$25
- Therefore, the marginal cost is  $(\$25/3)$ , which equals \$8.33

# Average Cost

Three **average costs** of production are

1. Average fixed cost
  2. Average variable cost
  3. Average total cost
- 
- **Average fixed cost (AFC)** is total fixed cost per unit of output
  - **Average variable cost (AVC)** is total variable cost per unit of output
  - **Average total cost (ATC)** is total cost per unit of output

# Average Cost

The average cost concepts are calculated from the total cost concepts as follows:

$$TC = TFC + TVC.$$

Divide each total cost term by the quantity produced,  
 $Q$ , to get

$$\frac{TC}{Q} = \frac{TFC}{Q} + \frac{TVC}{Q},$$

or

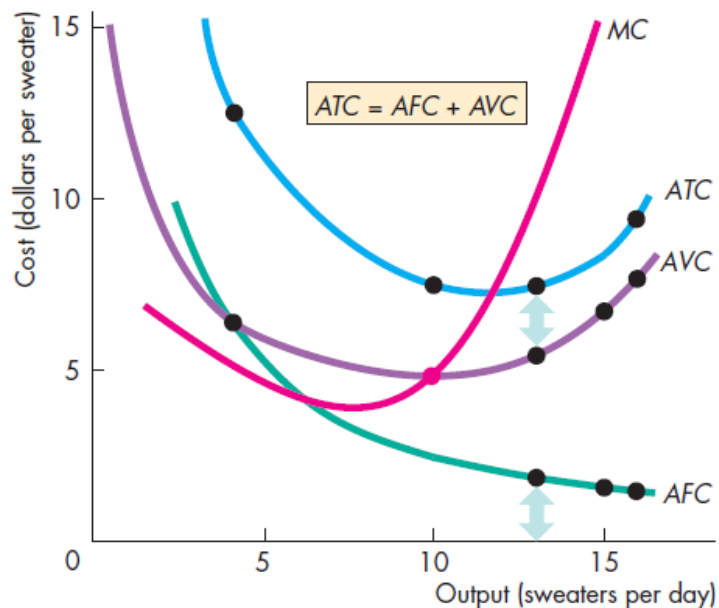
$$ATC = AFC + AVC.$$

# Costs

|   | Labor<br>(workers<br>per day) | Output<br>(sweaters<br>per day) | Total<br>fixed<br>cost<br>(TFC) | Total<br>variable<br>cost<br>(TVC) | Total<br>cost<br>(TC) | Marginal<br>cost<br>(MC)            | Average<br>fixed<br>cost<br>(AFC) | Average<br>variable<br>cost<br>(AVC) | Average<br>total<br>cost<br>(ATC) |
|---|-------------------------------|---------------------------------|---------------------------------|------------------------------------|-----------------------|-------------------------------------|-----------------------------------|--------------------------------------|-----------------------------------|
|   |                               |                                 | (dollars per day)               |                                    |                       | (dollars per<br>additional sweater) | (dollars per sweater)             |                                      |                                   |
| A | 0                             | 0                               | 25                              | 0                                  | 25                    | ..... 6.25                          | —                                 | —                                    | —                                 |
| B | 1                             | 4                               | 25                              | 25                                 | 50                    | ..... 4.17 $(75-50)/(10-4)$         | 6.25                              | 6.25                                 | 12.50                             |
| C | 2                             | 10                              | 25                              | 50                                 | 75                    | ..... 8.33                          | 2.50 $25/10$                      | 5.00 $50/10$                         | 7.50                              |
| D | 3                             | 13                              | 25                              | 75                                 | 100                   | ..... 12.50 $(125-100)/(15-13)$     | 1.92                              | 5.77                                 | 7.69                              |
| E | 4                             | 15                              | 25                              | 100                                | 125                   | ..... 25.00                         | 1.67 $25/15$                      | 6.67 $100/15$                        | 8.33                              |
| F | 5                             | 16                              | 25                              | 125                                | 150                   |                                     | 1.56                              | 7.81                                 | 9.38                              |

# Marginal Cost and Average Cost

**FIGURE 11.5** Marginal Cost and Average Costs



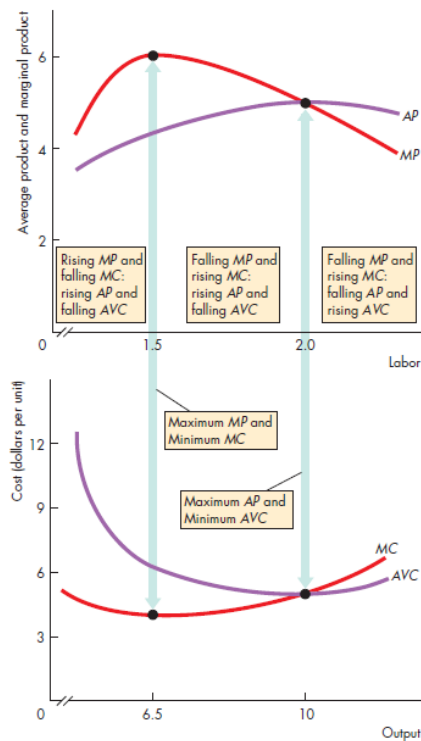
- MC curve is U-shaped - MC decreases at low levels of output due to increasing MP and eventually MC cost rises due to diminishing MP
- MC curve intersects the AVC curve and the ATC curve at their minimum points
- The average fixed cost curve (AFC) is downward sloping [as output increases, the same constant TFC is spread over a larger output]
- The ATC curve and AVC curve are U-shaped
- The vertical distance between the ATC curve and the AVC curve is equal to AFC (average fixed cost)

# Marginal Cost and Average Cost

- The marginal cost (MC) curve intersects the average variable cost curve and the average total cost curve at their minimum points
- When marginal cost is less than average cost, average cost is decreasing, and when marginal cost exceeds average cost, average cost is increasing
- This relationship holds for both the ATC curve and the AVC curve

# Cost Curves and Product Curves

FIGURE 11.6 Product Curves and Cost Curves



# Shifts in the Cost Curve

The position of a firm's short-run cost curves depends on two factors:

- Technology
- Prices of factors of production

# Shifts in the Cost Curve

## ■ Technology

- A technological change that increases productivity increases the marginal product and average product of labour
- With a better technology, the same factors of production can produce more output, so the technological advancement lowers the costs of production and shifts the cost curves downward

# Shifts in the Cost Curve

## ■ Prices of factors of production

- An increase in the price of a factor of production increases the firm's costs and shifts its cost curves. How the curves shift depends on which factor price changes
- An increase in rent or some other component of fixed cost shifts the TFC, AFC and TC curves upward but leaves the AVC, TVC and MC curves unchanged. Example: if the interest expense paid by a trucking company increases
- An increase in wages, gasoline, or another component of variable cost shifts the TVC, AVC and MC curves upward but leaves the AFC and TFC curves unchanged

# A Compact Glossary of Costs

| Term                   | Symbol | Definition                                                                         | Equation                       |
|------------------------|--------|------------------------------------------------------------------------------------|--------------------------------|
| Fixed cost             |        | Cost that is independent of the output level; cost of a fixed factor of production |                                |
| Variable cost          |        | Cost that varies with the output level; cost of a variable factor of production    |                                |
| Total fixed cost       | $TFC$  | Cost of the fixed factors of production                                            |                                |
| Total variable cost    | $TVC$  | Cost of the variable factors of production                                         |                                |
| Total cost             | $TC$   | Cost of all factors of production                                                  | $TC = TFC + TVC$               |
| Output (total product) | $TP$   | Total quantity produced (output $Q$ )                                              |                                |
| Marginal cost          | $MC$   | Change in total cost resulting from a one-unit increase in total product           | $MC = \Delta TC \div \Delta Q$ |
| Average fixed cost     | $AFC$  | Total fixed cost per unit of output                                                | $AFC = TFC \div Q$             |
| Average variable cost  | $AVC$  | Total variable cost per unit of output                                             | $AVC = TVC \div Q$             |
| Average total cost     | $ATC$  | Total cost per unit of output                                                      | $ATC = AFC + AVC$              |

# Long-Run Cost

- In the long run, a firm can vary both the quantity of labour and the quantity of capital, so in the long run, all the firm's costs are variable
- The behavior of long-run cost depends on the firm's production function, which is the relationship between the maximum output attainable and the quantities of both labour and capital

# The Production Function

Production function is the relationship between quantity of inputs used to make a good and the quantity of output of that good

**TABLE 11.3** The Production Function

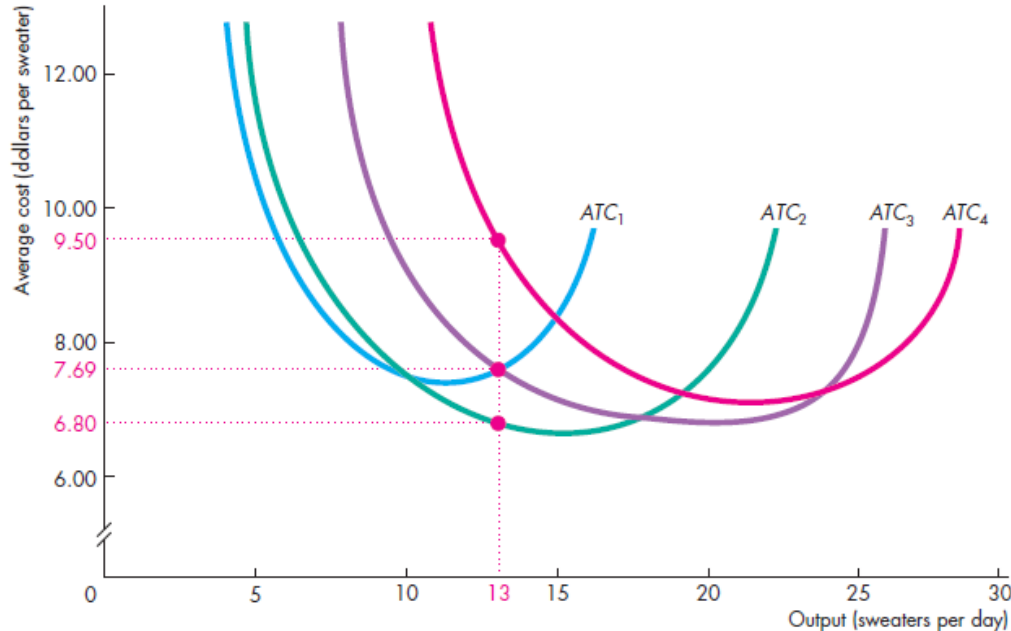
| Labor<br>(workers per day)    | Output<br>(sweaters per day) |         |         |         |
|-------------------------------|------------------------------|---------|---------|---------|
|                               | Plant 1                      | Plant 2 | Plant 3 | Plant 4 |
| 1                             | 4                            | 10      | 13      | 15      |
| 2                             | 10                           | 15      | 18      | 20      |
| 3                             | 13                           | 18      | 22      | 24      |
| 4                             | 15                           | 20      | 24      | 26      |
| 5                             | 16                           | 21      | 25      | 27      |
| Knitting machines<br>(number) | 1                            | 2       | 3       | 4       |

Diminishing returns:

- For a given plant size, the marginal product of labour diminishes as more labour is employed (**diminishing marginal product of labour**)
- For a given quantity of labour, the marginal product of capital diminishes as the quantity of capital used increases (**diminishing marginal product of capital**)
- The MP of capital is the change in TP divided by the change in capital when the quantity of labour is constant—i.e, the change in output resulting from a one-unit increase in the quantity of capital

# Short-Run Cost and Long-Run Cost

FIGURE 11.7 Short-Run Costs of Four Different Plants



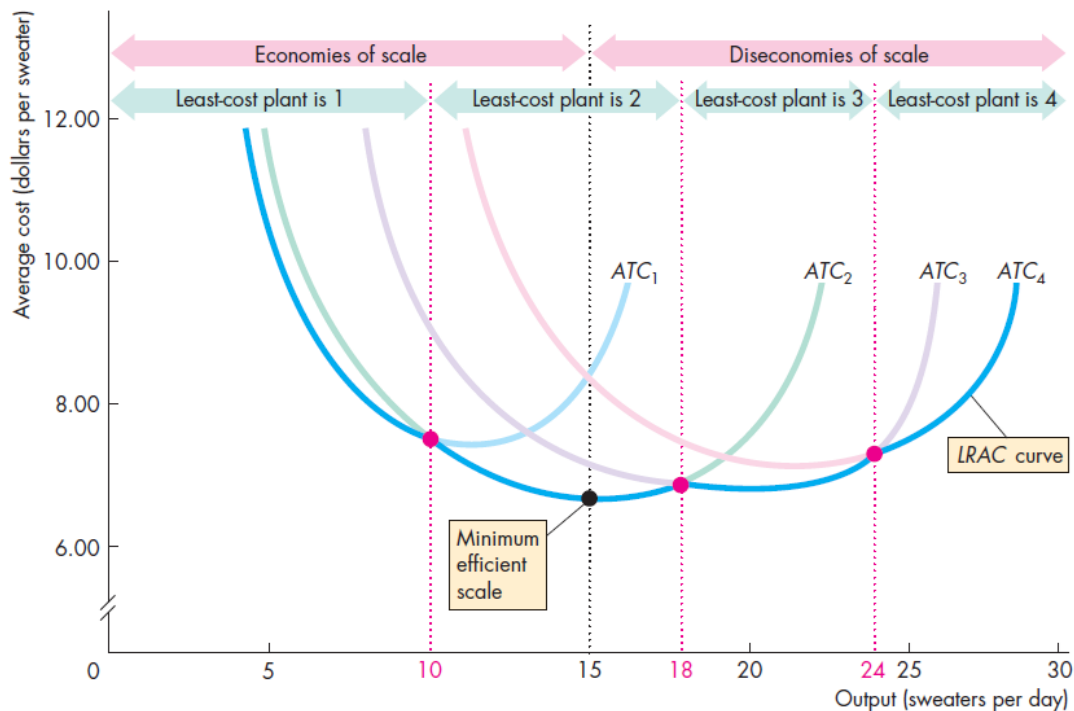
- Each short-run ATC curve is U-shaped because, as the quantity of labour increases, its marginal product initially increases and then diminishes
- The economically efficient plant for producing a given output is the one that has the lowest average total cost - which is to produce 13 sweaters a day is the one with 2 machines

# Long-Run Average Cost Curve

- When a firm is producing a given output at the least possible cost, it is operating on its long-run average cost curve
- The long-run average cost (LRAC) curve is the relationship between the lowest attainable average total cost and output when the firm can change both the plant it uses and the quantity of labour it employs
- LRAC curve comprises of the combination of the cheapest SRAC curves
- The long-run average cost curve is a planning curve. It tells the firm, the plant and the quantity of labour to use at each output to minimize average cost

# Long-Run Average Cost Curve

**FIGURE 11.8** Long-Run Average Cost Curve



# Economies and Diseconomies of Scale

- **Economies of scale** are features of a firm's technology that make average total cost fall as output increases
- When economies of scale are present, the LRAC curve slopes downward
- In other words, when long-run average total cost declines as output increases, there are said to be economies of scale
- In Fig. 11.8 (previous slide), the firm has economies of scale for outputs up to 15 sweaters a day
- Greater specialization of both labour and capital is the main source of economies of scale

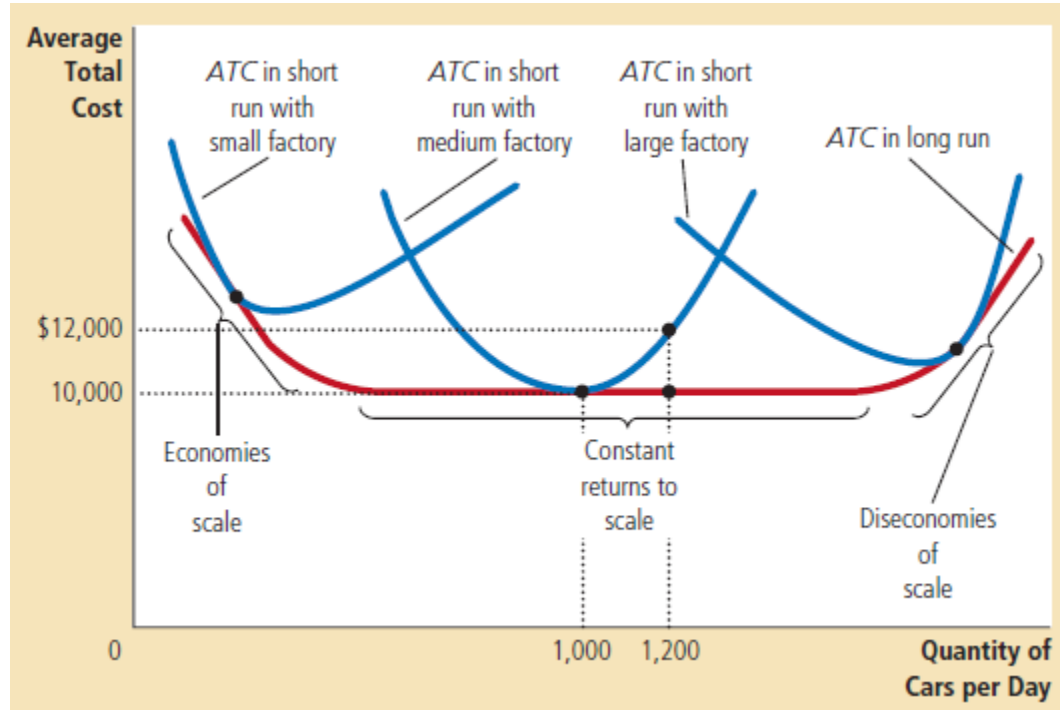
# Economies and Diseconomies of Scale

- **Diseconomies of scale** are features of a firm's technology that make average total cost rise as output increases
- When diseconomies of scale are present, the LRAC curve slopes upward
- In other words, when long-run average total cost rises as output increases, there are said to be diseconomies of scale
- In Fig. 11.8, the firm experiences diseconomies of scale at outputs greater than 15 sweaters a day

# Constant Returns to Scale and Minimum Efficient Scale

- **Constant returns to scale** are features of a firm's technology that keep average total cost constant as output increases
- When constant returns to scale are present, the LRAC curve is horizontal
- A firm's **minimum efficient scale** is the smallest output at which long-run average cost reaches its lowest level

# Average Total Cost in the Short and Long Runs



# Exercise

| Output (Q) | Fixed Cost (FC) | Variable Cost (VC) | Total Cost (TC) | Average Fixed Cost (AFC) | Average Variable Cost (AVC) | Average Total Cost (ATC) | Marginal Cost (MC) |
|------------|-----------------|--------------------|-----------------|--------------------------|-----------------------------|--------------------------|--------------------|
| 0          |                 | 0                  | \$800           | -                        | -                           | -                        | -                  |
| 200        |                 | 650                |                 |                          |                             |                          |                    |
| 450        |                 |                    |                 | 1.78                     | 2.89                        |                          |                    |
| 550        |                 |                    | 2,750           |                          |                             | 5                        | 6.5                |
| 600        |                 | 2,600              | 3,400           | 1.33                     |                             | 5.67                     |                    |
| 625        |                 |                    |                 |                          | 5.2                         |                          |                    |
| 640        |                 |                    |                 |                          |                             | 7.34                     | 43.33              |